In [1]:
```
# DataFlair Iris Flower Classification
# Import Packages
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.neighbors import KNeighborsClassifier
```

In [2]:
```
#columns = ['Sepal length', 'Sepal width', 'Petal length', 'Petal width', 'Class_labels']
# Load the data
df = pd.read_csv('iris.csv')
print(df.head(5))
print(df.columns)
```

```
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
       'species'],
      dtype='object')
```

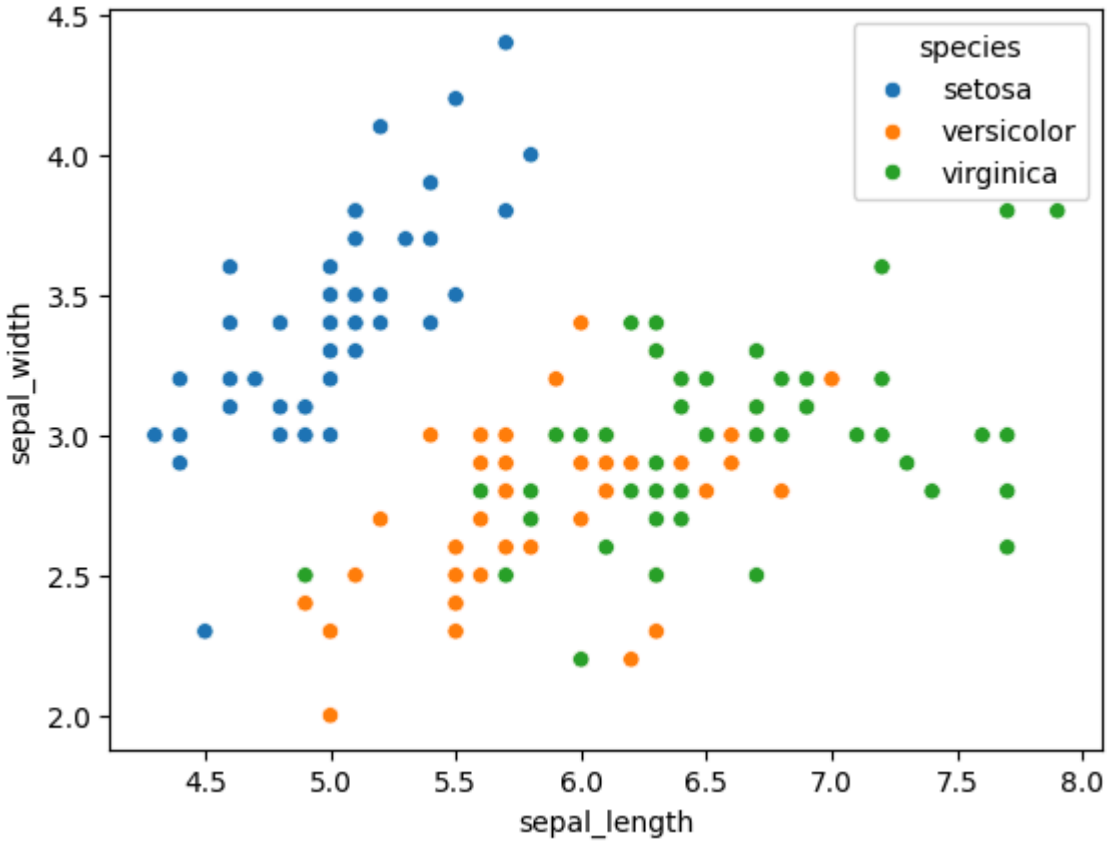
In [3]:
```
print(df.describe())
df['species'].unique()
```

```
       sepal_length  sepal_width  petal_length  petal_width
count    150.000000   150.000000    150.000000   150.000000
mean       5.843333     3.054000      3.758667     1.198000
std        0.828066     0.433594      1.764420     0.763115
min        4.300000     2.000000      1.000000     0.100000
25%        5.100000     2.800000      1.600000     0.300000
50%        5.800000     3.000000      4.350000     1.300000
75%        6.400000     3.300000      5.100000     1.800000
max        7.900000     4.400000      6.900000     2.500000
```

Out[3]:
```
array(['setosa', 'versicolor', 'virginica'], dtype=object)
```

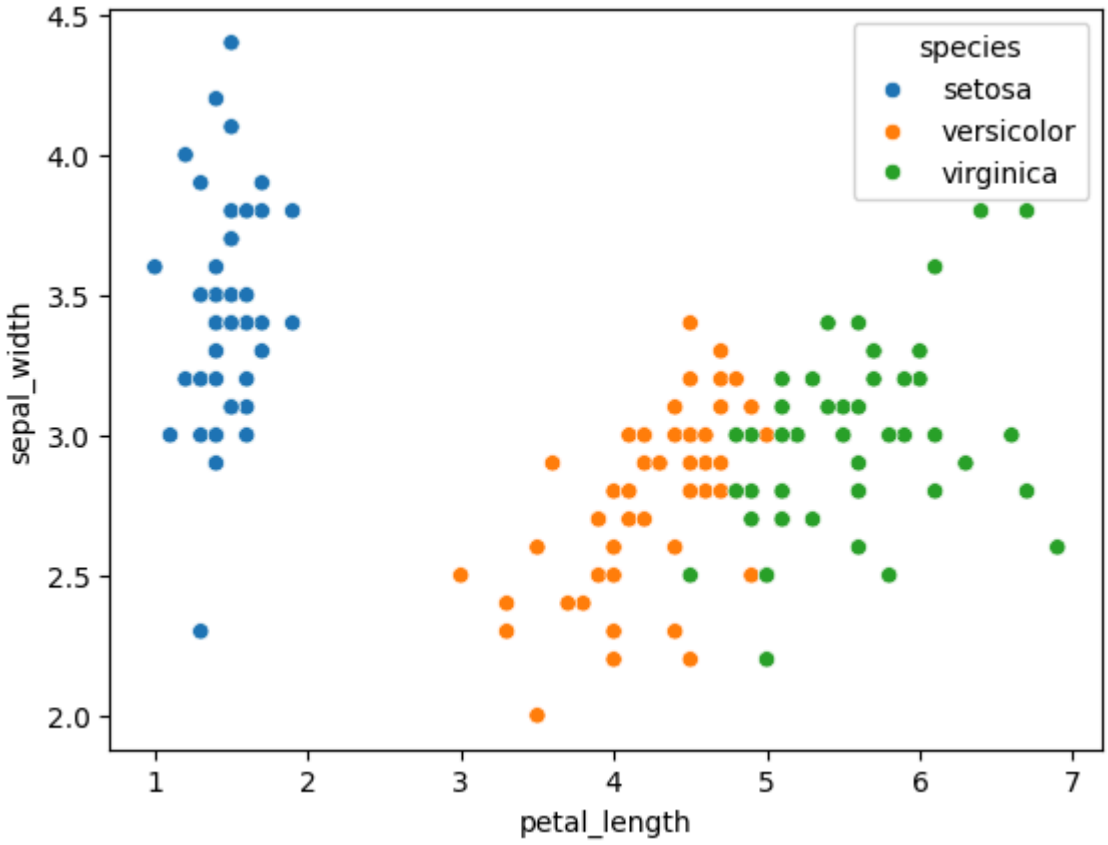
In [4]:
```
sns.scatterplot(x=df['sepal_length'], y=df['sepal_width'], hue=df['species'], data=df)
```

Out[4]:
```
<Axes: xlabel='sepal_length', ylabel='sepal_width'>
```



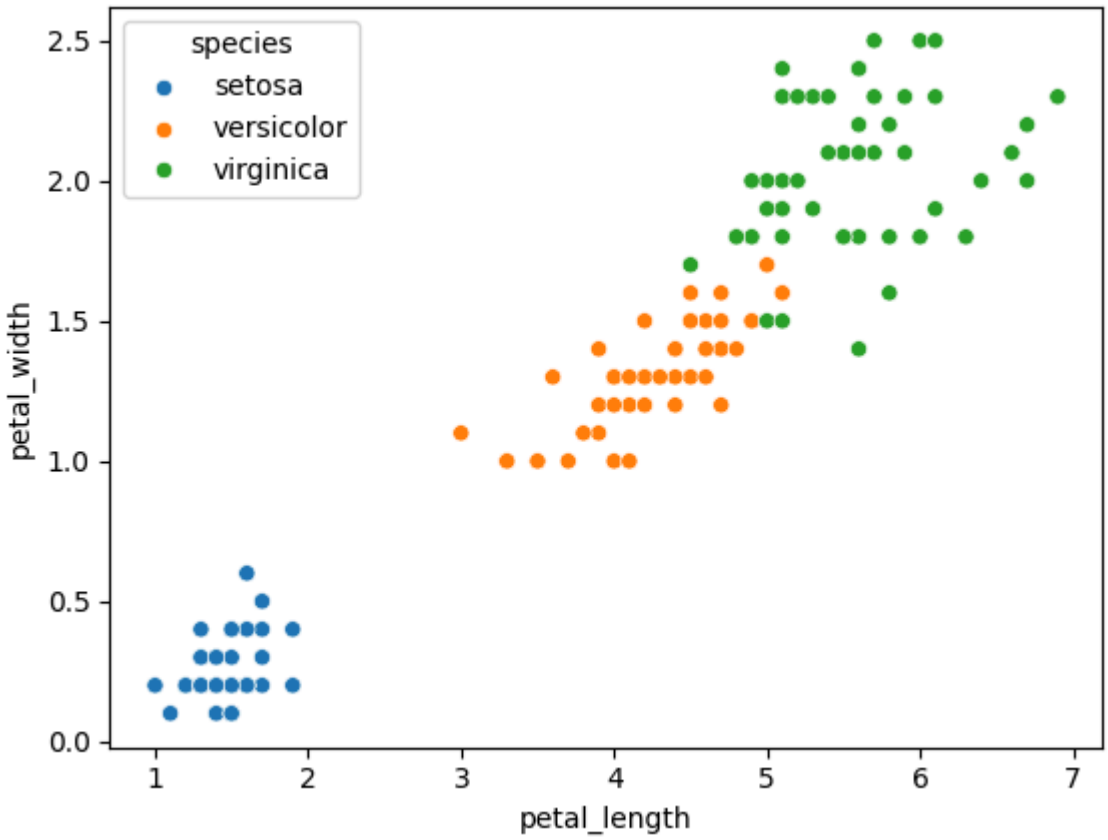
In [5]:
```
sns.scatterplot(x=df['petal_length'], y=df['sepal_width'], hue=df['species'], data=df)
```

Out[5]: <Axes: xlabel='petal_length', ylabel='sepal_width'>

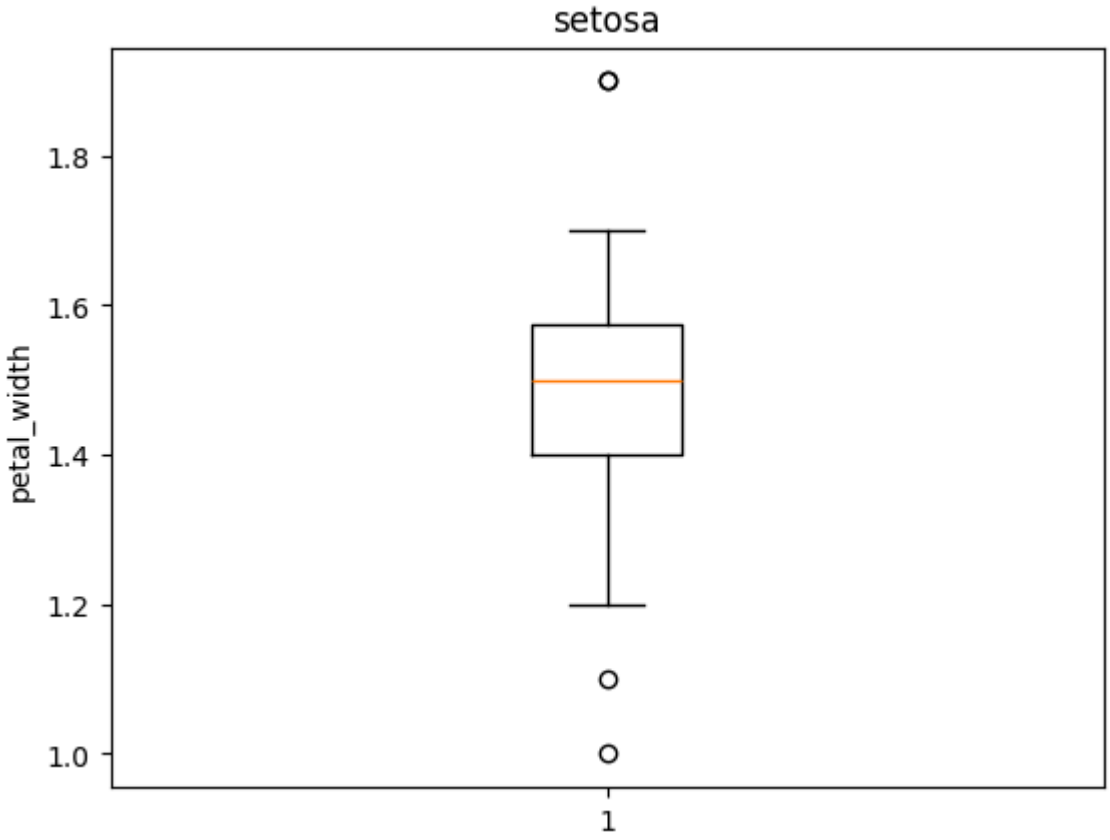


In [6]:
```
sns.scatterplot(x=df['petal_length'], y=df['petal_width'], hue=df['species'], data=df)
```

Out[6]: <Axes: xlabel='petal_length', ylabel='petal_width'>



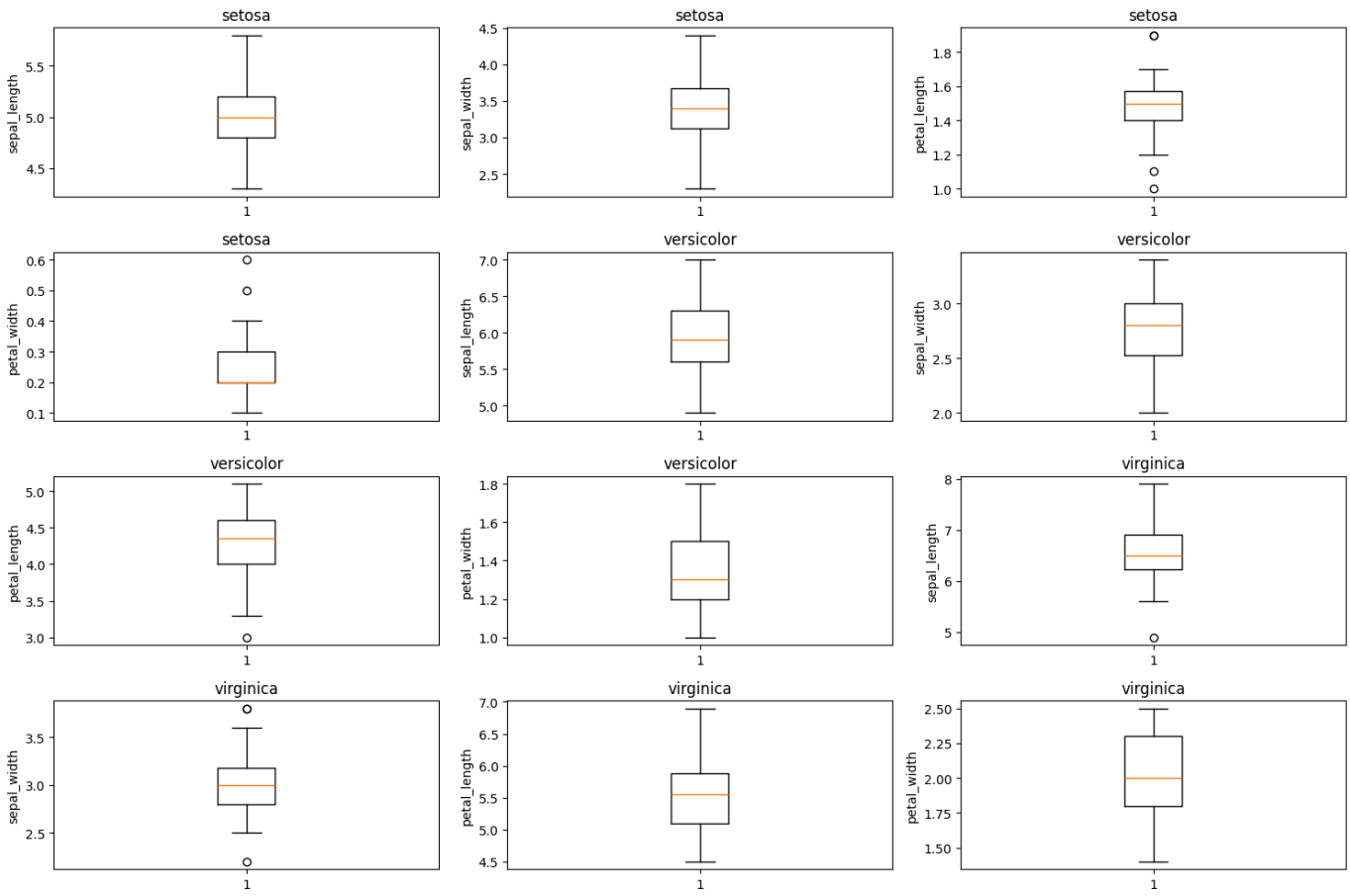
In [7]:
```
setosa_data = df[df['species'] == 'setosa']
plt.boxplot(setosa_data['petal_length'])
plt.title('setosa')
plt.ylabel('petal_width')
plt.show()
```



In [8]:
```
#'Iris-setosa', 'Iris-versicolor', 'Iris-virginica'
fig, axes = plt.subplots(4, 3, figsize=(15, 10))
axes = axes.flatten()
print(axes)
setosa_data = df[df['species'] == 'setosa']
axes[0].boxplot(setosa_data['sepal_length'])
axes[0].set_title('setosa')
```

```
axes[0].set_ylabel('sepal_length')
setosa_data = df[df['species'] == 'setosa']
axes[1].boxplot(setosa_data['sepal_width'])
axes[1].set_title('setosa')
axes[1].set_ylabel('sepal_width')
setosa_data = df[df['species'] == 'setosa']
axes[2].boxplot(setosa_data['petal_length'])
axes[2].set_title('setosa')
axes[2].set_ylabel('petal_length')
setosa_data = df[df['species'] == 'setosa']
axes[3].boxplot(setosa_data['petal_width'])
axes[3].set_title('setosa')
axes[3].set_ylabel('petal_width')
versicolor_data = df[df['species'] == 'versicolor']
axes[4].boxplot(versicolor_data['sepal_length'])
axes[4].set_title('versicolor')
axes[4].set_ylabel('sepal_length')
versicolor_data = df[df['species'] == 'versicolor']
axes[5].boxplot(versicolor_data['sepal_width'])
axes[5].set_title('versicolor')
axes[5].set_ylabel('sepal_width')
versicolor_data = df[df['species'] == 'versicolor']
axes[6].boxplot(versicolor_data['petal_length'])
axes[6].set_title('versicolor')
axes[6].set_ylabel('petal_length')
versicolor_data = df[df['species'] == 'versicolor']
axes[7].boxplot(versicolor_data['petal_width'])
axes[7].set_title('versicolor')
axes[7].set_ylabel('petal_width')
verginica_data = df[df['species'] == 'virginica']
axes[8].boxplot(verginica_data['sepal_length'])
axes[8].set_title('virginica')
axes[8].set_ylabel('sepal_length')
verginica_data = df[df['species'] == 'virginica']
axes[9].boxplot(verginica_data['sepal_width'])
axes[9].set_title('virginica')
axes[9].set_ylabel('sepal_width')
verginica_data = df[df['species'] == 'virginica']
axes[10].boxplot(verginica_data['petal_length'])
axes[10].set_title('virginica')
axes[10].set_ylabel('petal_length')
verginica_data = df[df['species'] == 'virginica']
axes[11].boxplot(verginica_data['petal_width'])
axes[11].set_title('virginica')
axes[11].set_ylabel('petal_width')
plt.tight_layout()
```

```
[<Axes: > <Axes: > <Axes: > <Axes: > <Axes: > <Axes: > <Axes: > <Axes: >
 <Axes: > <Axes: > <Axes: > <Axes: >]
```



In [9]:
```
for flower in [setosa_data,versicolor_data,verginica_data]:
    for dimension in ['sepal_length','sepal_width','petal_length','petal_width']:
        Q1 = flower[dimension].quantile(0.25)
        Q3 = flower[dimension].quantile(0.75)
        IQR = Q3 - Q1
        # Determine outliers using the IQR method
        outliers = flower[(flower[dimension] < Q1 - 1.5 * IQR) | (flower[dimension] > Q3 + 1.
        print(outliers)
        df = df.drop(outliers.index)
```

```
Empty DataFrame
Columns: [sepal_length, sepal_width, petal_length, petal_width, species]
Index: []
Empty DataFrame
Columns: [sepal_length, sepal_width, petal_length, petal_width, species]
Index: []
    sepal_length  sepal_width  petal_length  petal_width species
13           4.3          3.0           1.1          0.1  setosa
22           4.6          3.6           1.0          0.2  setosa
24           4.8          3.4           1.9          0.2  setosa
44           5.1          3.8           1.9          0.4  setosa
    sepal_length  sepal_width  petal_length  petal_width species
23           5.1          3.3           1.7          0.5  setosa
43           5.0          3.5           1.6          0.6  setosa
Empty DataFrame
Columns: [sepal_length, sepal_width, petal_length, petal_width, species]
Index: []
Empty DataFrame
Columns: [sepal_length, sepal_width, petal_length, petal_width, species]
Index: []
    sepal_length  sepal_width  petal_length  petal_width     species
98           5.1          2.5           3.0          1.1  versicolor
Empty DataFrame
Columns: [sepal_length, sepal_width, petal_length, petal_width, species]
Index: []
     sepal_length  sepal_width  petal_length  petal_width    species
106           4.9          2.5           4.5          1.7  virginica
     sepal_length  sepal_width  petal_length  petal_width    species
117           7.7          3.8           6.7          2.2  virginica
119           6.0          2.2           5.0          1.5  virginica
131           7.9          3.8           6.4          2.0  virginica
Empty DataFrame
Columns: [sepal_length, sepal_width, petal_length, petal_width, species]
Index: []
Empty DataFrame
Columns: [sepal_length, sepal_width, petal_length, petal_width, species]
Index: []
```

In [10]:
```
df.describe()
```

Out[10]:

	sepal_length	sepal_width	petal_length	petal_width
count	139.000000	139.000000	139.000000	139.000000
mean	5.870504	3.041007	3.805755	1.217266
std	0.789832	0.419595	1.733672	0.761570
min	4.400000	2.000000	1.200000	0.100000
25%	5.200000	2.800000	1.550000	0.300000
50%	5.800000	3.000000	4.400000	1.300000
75%	6.400000	3.250000	5.100000	1.800000
max	7.700000	4.400000	6.900000	2.500000

In [11]:
```
data = df.values
#print(data[0:6])
X = data[:,0:4]
6.900000
2.500000
Y = data[:,4]
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.3)
```

In [12]:
```
#svn = SVC()
#svn.fit(X_train, y_train)
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, Y_train)
```

Out[12]:

KNeighborsClassifier

▶ Parameters

In [13]:
```
#predictions = svn.predict(X_test)
predictions = knn.predict(X_test)
```

In [14]:
```
accuracy_score(Y_test, predictions)
```

Out[14]: 0.9761904761904762

In [15]:
```
X_new = np.array([[5.1,4.0,1.4,0.2]])
#Prediction of the species from the input vector
prediction = knn.predict(X_new)
print("Prediction of Species:", prediction)
```

```
Prediction of Species: ['setosa']
```

In [ ]: